

ADDITIONAL QUESTIONS

LIMITS, CONTINUITY,
DIFFERENTIABILITY

SINGLE CORRECT ANSWER TYPE

1. Let $f(x) = \lim_{n \rightarrow \infty} \sum_{r=1}^n 3^{r-1} \sin^3 \frac{x}{3^r}$ and $g(x) = x - 4f(x)$ then $\lim_{x \rightarrow 0} (1 + g(x))^{\cot x}$ is
 A) $\frac{1}{e}$ B) e^2 C) e D) 1
2. Let $f(x)$ be a real valued function defined on the interval $(0, \infty)$ by

$$f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} \, dt$$
 Then which of the following statements is (are) true

[IIT-2010]

- (A) $f^{(1)}(x)$ exists for all $x \in (0, \infty)$
 (B) $f'(x)$ exists for all $x \in (0, \infty)$ and $f'(x)$ is continuous on $(0, \infty)$ but not differentiable on $(0, \infty)$
 (C) There exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
 (D) There exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$
3. Let $f(x) = \text{degree of } (u^{x^2} + u^2 + 2u - 3)$ (where u is a real parameter, x is variable) then
 A) $f(x)$ is continuous at $x = \sqrt{2}$ but not differentiable
 B) $f(x)$ is neither differentiable nor continuous at $x = \sqrt{2}$

C) $f(x)$ is discontinuous at $x = \sqrt{2}$ D) $f(x)$ is differentiable at $x = \sqrt{2}$

4. Given

$$f(x) = \begin{cases} \log_a (a \lfloor x \rfloor + \lceil -x \rceil)^x \left(\frac{a^{\frac{2}{\lfloor x \rfloor + \lceil -x \rceil}}}{3 + a^{\frac{1}{\lfloor x \rfloor}}} \right)^{-5} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

 $|x| \neq 0, a > 1$ and $f(x) = 0$ for $x = 0$ (where $\lfloor \cdot \rfloor$ is G.I.F) then

- A) f is continuous but not differentiable at $x = 0$
 B) f is continuous and differentiable at $x = 0$
 C) The differentiability of f at $x = 0$ depends on the value of a
 D) f is continuous and differentiable at $x = 0$ and for $a = e$ only
5. Let $f(0) = 0$ and $f'(0) = 1$. For a positive integer K , $\lim_{x \rightarrow 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + \dots + f\left(\frac{x}{k}\right) \right)$ is
 A) $\frac{k(k+1)}{2}$ B) $\frac{k+1}{2k^2}$
 C) $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$ D) $\frac{1}{2}$
6. Let $f: [a, b] \rightarrow [1, \infty)$ be a continuous function and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0, & \text{if } x < a \\ \int_a^x f(t) dt, & \text{if } a \leq x \leq b \\ \int_a^b f(t) dt, & \text{if } x > b \end{cases}$$

- (A) $g(x)$ is continuous but not differentiable at a
 (B) $g(x)$ is differentiable on $\mathbb{R}$
 (C) $g(x)$ is continuous but not differentiable at b
 (D) $g(x)$ is continuous and differentiable at either a or b but not both

MULTI ANSWER QUESTIONS

7. Which of the following statement(s) is/are TRUE?

Statement A: If function $y = f(x)$ is continuous at $x = c$ such that $f(c) \neq 0$ then $f(x)f(c) > 0 \forall x \in (c-h, c+h)$ where h is sufficiently small positive quantity.

Statement B:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left[\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right] = 1 + 2 \ln 2$$

Statement C: Let f be a continuous and non-negative function defined on $[a, b]$. if

$$\int_a^b f(x) dx = 0 \text{ then } f(x) = 0 \forall x \in [a, b].$$

Statement D: Let f be a continuous function defined on $[a, b]$ such that

$$\int_a^b f(x) dx = 0, \text{ then there exists atleast}$$

one $c \in (a, b)$ for which $f(c) = 0$.

8. If $a, b, c \in \mathbb{R}^+$ then $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(k+an)(k+bn)}$ is equal to

A) $\frac{1}{a-b} \ln \frac{b(b+1)}{a(a+1)}$ if $a \neq b$

B) $\frac{1}{a-b} \ln \frac{a(b+1)}{b(a+1)}$ if $a \neq b$

C) non existent if $a = b$

D) equals $\frac{1}{a(1+a)}$ if $a = b$

9. Let $f(x) = \begin{cases} \int_0^x (1+|1-t|) dt & , x > 2 \\ 5x+1 & , x \leq 2 \end{cases}$, then

(A) $f(x)$ is not continuous at $x = 2$

(B) $f(x)$ is continuous but not differentiable at $x = 2$

(C) $f(x)$ is differentiable everywhere

(D) the right derivative of $f(x)$ at $x = 2$ does not exist

10. If $f'(x) = f(x) + \int_{-10}^{10} f(x) dx$, then

(A) $f'(x) = f(x)$ if $\lim_{x \rightarrow \infty} f(x) = 0$

(B) $f'(x) - f(x) = c$ (any constant)

(C) $f''(x) - f(x) = c$ (any constant)

(D) None of the above

11. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$

such that $f(x) = f(1-x)$ and $f'\left(\frac{1}{4}\right) = 0$.

Then, [IIT-2008]

(A) $f''(x)$ vanishes at least twice on $[0, 1]$

(B) $f'\left(\frac{1}{2}\right) = 0$

(C) $\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right) \sin x dx = 0$

(D) $\int_0^{1/2} f(t) e^{\sin \pi t} dt = \int_{1/2}^1 f(1-t) e^{\sin \pi t} dt$

COMPREHENSION TYPE

Passage : L' Hospital's rule has many versions one them is this suppose $f, g : (a, b) \rightarrow \mathbb{R}$ are differentiable on (a, b) . Suppose further that

i) $g^1(x) \neq 0$ for $x \in (a, b)$

(ii) $\lim_{x \rightarrow a^+} g(x) \rightarrow \infty$ (or $-\infty$)

(iii) $\lim_{x \rightarrow a^+} \frac{f^1(x)}{g^1(x)} = L$ then $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L$

(This rule can be extended to cover the case where a and b tends to infinity or L tends to infinity)

12. Let 'f' be a differentiable function on $(0, \infty)$.

If $\lim_{x \rightarrow \infty} \left(\sin\left(\frac{\pi}{10}\right) f(x) + f^1(x) \right) = \sec\left(\frac{\pi}{5}\right)$ then $\lim_{x \rightarrow \infty} f(x)$ equals

- A) $\frac{1}{4}$ B) 4 C) $3 - \sqrt{5}$ D) $\frac{3 + \sqrt{5}}{4}$

13. Let 'f' be a differentiable function on $(0, \infty)$.

If $\lim_{x \rightarrow \infty} \left(\tan\left(\frac{\pi}{8}\right) f(x) + 2\sqrt{x} f^1(x) \right) = \cot \frac{\pi}{12}$ then $\lim_{x \rightarrow \infty} f(x)$ equals.

- A) $\sqrt{8} - \sqrt{6} + \sqrt{4} - \sqrt{3}$
 B) $\sqrt{8} + \sqrt{6} - \sqrt{4} - \sqrt{3}$
 C) $\sqrt{3} + \sqrt{4} + \sqrt{6} + \sqrt{8}$
 D) $\sqrt{8} - \sqrt{6} - \sqrt{4} + \sqrt{3}$

Passage : Let 'f' be a polynomial function satisfying

$$f(x).f(y) + 2 = f(x) + f(y) + f(xy) \quad \forall x, y \in \mathbb{R}^+ \cup \{0\} \text{ and}$$

f(x) is one-one with $f(0) = 1$ and $f^1(1) = 2$ then answer the following

14. The function $y = f(x)$ is given by

- A) $x^2 - 1$ B) $x^{1/3} - 1$
 C) $1 + \frac{2x^3}{3}$ D) $1 + x^2$

15. The number of points of non-differentiability

of $h(x) = \min \left\{ \frac{2}{f(x)}, x^2, |1 - |x|| \right\}, x \geq 0$ is

- A) 6 B) 5 C) 3 D) 4

MATCHING TYPE QUESTIONS

16. Match the following :

Column I

A)

$$f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - \{x\}^2) \right) \left(\sin^{-1}(1 - \{x\}) \right)}{\sqrt{2}(\{x\} - \{x\}^3)} & x > 0 \\ k & x = 0 \\ \frac{A \sin^{-1}(1 - \{x\}) \cos^{-1}(1 - \{x\})}{\sqrt{2}\{x\}(1 - \{x\})} & x < 0 \end{cases}$$

is continuous at $x = 0$ then the value of

$$\sin^2 k + \cos^2 \left(\frac{A\pi}{\sqrt{2}} \right) \text{ is}$$

B) $f(x) = [2 + 5|n| \sin x] n \in \mathbb{Z}$ has exactly 19 points of non

differentiability in $x \in (0, \pi)$ then possible values of n are

$$C) \text{ If } f(x) = \begin{cases} x \cdot \frac{(3/4)^{1/x} - (3/4)^{-1/x}}{(3/4)^{1/x} - (3/4)^{-1/x}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

if $P = f'(0^-) - f'(0^+)$ then

$$\lim_{x \rightarrow P^-} \frac{(\exp(x+2) \ln 4)^{\frac{[x+1]}{4}} - 16}{4^x - 16} \text{ is } \leq$$

Column II

- P) 2
 Q) -2
 R) 3
 S) -3

INTEGER TYPE

17. Number of points where the function $f(x) = \left(x + \left[x^3 + 1\right]\right)^{x^2 + \sin x}$ is not differentiable for $x \in (1, 3/2)$ is (where $[.]$ is G.I.F)
18. Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the functional equation $f(xy) = e^{xy-x-y} (e^y f(x) + e^x f(y))$ $\forall x, y \in \mathbb{R}^+$ and $f(1) = e$ then $[\ln(f(e))]$ (where $[.]$ is G.I.F) is
19. Let $I_n = \int_{-1}^1 |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{2n}}{2n}\right) dx$. If $\lim_{n \rightarrow \infty} I_n$ can be expressed as rational $\frac{p}{q}$ in the lowest form, then the value of pq is

SUBJECTIVE QUESTIONS

20. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max\{f(t); 0 \leq t \leq x\}, & 0 \leq x \leq 1 \\ 3 - x; & 1 < x \leq 2 \end{cases}$. Discuss the continuity and differentiability of the function $g(x)$ in the interval $(0, 2)$.
21. Show that the transformation $z = \log \tan(x/2)$ reduces the differential equation $\frac{d^2 y}{dx^2} + \cot x \frac{dy}{dx} + 4y \csc^2 x = 0$ into $\frac{d^2 y}{dz^2} + 4y = 0$.
22. Let $f(x) = \begin{cases} 1+x & 0 \leq x \leq 2 \\ 3-x & 2 < x \leq 3 \end{cases}$. Determine $g(x) = f(f(x))$ and hence find the points of discontinuity of g , if any.
23. Let $f(x)$ be defined in the interval $[-2, 2]$ such that $f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$ and $g(x) = f(|x|) + |f(x)|$. Test the differentiability of $g(x)$ in $(-2, 2)$.
24. Let $\mathbb{R}$ be the set of real numbers and

$f: \mathbb{R} \rightarrow \mathbb{R}$ be such that for all x and y in $\mathbb{R}$, $|f(x) - f(y)| \leq |x - y|^3$. Prove that $f(x)$ is a constant.

25. Draw a graph of the function, $y = [x] + |1 - x|$, $-1 \leq x \leq 3$. Determine the points, if any, where this function is not differentiable, where $[.]$ denotes the greatest integer function.
26. Let $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. If the function $f(x)$ is continuous at $x = 1$, show that $f(x)$ is continuous for all $x \neq 0$.
27. If $f\left(\frac{x+y}{3}\right) = \frac{2+f(x)+f(y)}{3}$ for all real x and y and $f'(0) = 2$, then determine $y = f(x)$.
28. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} \max\{f(x), g(x)\}, & \text{if } x \leq 0 \\ \min\{f(x), g(x)\}, & \text{if } x > 0 \end{cases}$$

The number of points at which $h(x)$ is not differentiable is

29. Let the variable x_n be determined by the following law of formation $x_0 = \sqrt{a}$ $x_1 = \sqrt{a + \sqrt{a}}$, $x_2 = \sqrt{a + \sqrt{a + \sqrt{a}}}$ $x_3 = \sqrt{a + \sqrt{a + \sqrt{a + \sqrt{a}}}}$, then find $\lim_{n \rightarrow \infty} x_n$.
30. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$ for all x, y in $\mathbb{R}$ and $f(x) \neq 0$ for any x in $\mathbb{R}$. Let the function be differentiable at $x = 0$ and $f'(0) = 2$. Show that $f'(x) = 2f(x)$ for all x in $\mathbb{R}$. Hence determine $f(x)$. [IIT-1990]
31. If $f(x) = \sqrt{|x-1|}$ and $g(x) = \sin x$ then calculate $(f \circ g)(x)$ and $(g \circ f)(x)$ and discuss the differentiability of $(g \circ f)(x)$ at $x = 1$. [IIT-92]
32. Let $f[(x+y)/2] = \{f(x) + f(y)\}/2$ for all real x and y . If $f'(0)$ exists and equals -1 and

$f(0) = 1$, find $f(2)$. [IIT - 95]

33. (i) $\lim_{x \rightarrow 0} \frac{2(\tan x - \sin x) - x^3}{x^5}$.

(ii) $\lim_{x \rightarrow 1} \frac{(\ln(1+x) - \ln 2)(3 \cdot 4^{x-1} - 3x)}{\left[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}\right] \cdot \sin(x-1)}$

[IIT - 97]

(iii) Through a point A on a circle, a chord AP is drawn and on the tangent at A a point T is taken such that $AT = AP$. If TP produced meet the diameter through A at Q then prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

34. Let $\alpha \in \mathbb{R}$. Prove that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at α if and only if there is a function $g: \mathbb{R} \rightarrow \mathbb{R}$ which is continuous at α and satisfies $f(x) - f(\alpha) = g(x)(x - \alpha)$ for all $x \in \mathbb{R}$. [IIT - 2001]

35. If a function $f: [-2a, 2a] \rightarrow \mathbb{R}$ is an odd function such that $f(x) = f(2a - x)$ for $x \in [a, 2a]$ and the L H D at $x = a$ is zero then find the L H D at $x = -a$. [IIT - 2003]

36. $f'(0)$ can be represented for any differentiable function $f(-1, 1) \rightarrow \mathbb{R}$ as

$f'(0) = \lim_{x \rightarrow \infty} n f\left(\frac{1}{n}\right)$, where $f(0) = 0$. Find

$\lim_{n \rightarrow \infty} \left[\frac{n+1}{\pi/2} \cos^{-1}\left(\frac{1}{n}\right) - n \right]$, given

$\cos^{-1}\left(\frac{1}{n}\right) \leq \frac{\pi}{2} \quad \forall n \in \mathbb{N}$.

37. If $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ and $g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$. If right hand derivative at $x = 0$ exists for $f(x)$, then find the derivative of $g(x)$ at $x = 0$.

38. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined

by $f(x) = x \left\{ 1 + \frac{1}{3} \sin(\ln x^2) \right\}; x \neq 0$ and

$f(0) = 0$ is everywhere continuous, but has no differential coefficient at the origin.

39. Let f be a function satisfying

$f(x+y) + \sqrt{6-f(y)} = f(x)f(y)$ and

$f(h) \rightarrow 6$ as $h \rightarrow 0$.

Discuss the continuity of f .

40. Let f be a function such that

$f(x+f(y)) = f(f(x)) + f(y) \quad \forall x, y \in \mathbb{R}$

and $f(h) = h$ for $0 < h < \varepsilon$ where $\varepsilon > 0$,

then determine $f'(x)$ and $f(x)$.

41. The value of

$\lim_{n \rightarrow \infty} \left(\frac{1}{2n+1} + \frac{1}{2n+3} + \frac{1}{2n+5} + \dots + \frac{1}{4n-1} \right)$ is

$\frac{a}{b} \ln c$ where $a, b, c \in \mathbb{N}$. Then the least value of $a + b + c$ is

KEY

- 1) C 2) B 3) A 4) B
5) C 6) D 7) A, C, D 8) B, D
9) A, D 10) A, B, C 11) A, B, C, D
12) B 13) C 14) D 15) C
16) A-P, B-P, Q, C-Q, R
17) 2 18) 2 19) 6

22. Continuous in $(0, 2)$ and differentiable in $(0, 2)$ except at $x = 1$

23. $g(x) = \begin{cases} 2+x & 0 \leq x \leq 1 \\ 2-x & 1 < x \leq 2 \\ 4-x & 2 < x \leq 3 \end{cases}$

discontinuity at $x = 1, 2$ & non differentiable at $x = 0, 1$

25. $\{0, 1, 2, 3\}$

27. $2x + 2$

28. $g(x)$ is continuous for all $x \in [0, 2]$

but $g(x)$ is not differentiable at $x = 1$

29. 3

30. e^{2x}

31. not differentiable at $x=1$

32. -1

33. (i) $1/4$, (ii) $-\frac{9}{4} \ln \frac{4}{e}$
34. $f(x)$ is differentiable at $x = \alpha$
If $g(x)$ is continuous at $x = \alpha$.
35. 0
36. $\frac{\pi - 2}{\pi}$
37. 0
39. If $f(x) \neq 0$, then f is discontinuous at x . If $f(x) = 0$, then $f(x)$ is continuous at x .
40. $f(x) = x$ & $f'(x) = 1$
41. 5

HINTS

1.
$$f(x) = \frac{1}{4} \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(3^r \sin \frac{x}{3^r} - 3^{r-1} \cdot \sin \frac{x}{3^{r-1}} \right)$$

$$= \frac{1}{4} \lim_{n \rightarrow \infty} \left(3 \sin \frac{x}{3} - \sin x + 3^2 \sin \frac{x}{3^2} - 3 \sin \frac{x}{3} + \dots + 3^n \sin \frac{x}{3^n} - 3^{n-1} \sin \frac{x}{3^{n-1}} \right)$$

$$= \frac{1}{4} \lim_{n \rightarrow \infty} \left(3^n \sin \frac{x}{3^n} - \sin x \right)$$

$$= \frac{1}{4} \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{x}{3^n}}{\left(\frac{x}{3^n} \right)} \cdot x - \sin x \right) = \frac{x - \sin x}{4}$$

$$g(x) = x - 4 + \sin x = \sin x$$

$$\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x} = e^{\lim_{x \rightarrow 0} \frac{\sin x \cdot \cos x}{\sin x}} = e$$
2. $f^1(x) = \frac{1}{x} + \sqrt{1 + \sin x}$ exists and continuous
 $\forall x \in (0, \infty)$

$$f^{11}(x) = -\frac{1}{x^2} + \frac{\cos x}{2\sqrt{1 + \sin x}}$$

 $f^1(x)$ is not differentiable at $\sin x = -1$
 where $x = 2n\pi - \frac{\pi}{2}$
 $f^1(x)$ is not differentiable at $\sin x = -1$

where $x = 2n\pi - \frac{\pi}{2}$

$f(x)$ and $f^1(x)$ are positive for all

$x \in (\pi, \infty)$

$$f(x) > f^1(x) \Rightarrow |f(x)| > |f^1(x)|$$

$\frac{1}{x}$ is not bounded hence D is not correct

3. $f(x) = 2$ if $x \leq \sqrt{2}$
 $= x^2$ if $x > \sqrt{2}$
 $\therefore \lim_{x \rightarrow \sqrt{2}^+} f(x) = \lim_{x \rightarrow \sqrt{2}^+} x^2 = 2, \lim_{x \rightarrow \sqrt{2}^-} f(x) = 2 \Rightarrow$ It is continuous at $x = \sqrt{2}$

$$f^1(\sqrt{2}^+) = \lim_{h \rightarrow 0^+} \frac{f(h + \sqrt{2}) - f(\sqrt{2})}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{(h + \sqrt{2})^2 - 2}{h} = 2\sqrt{2}$$

$$f^1(\sqrt{2}^-) = \lim_{h \rightarrow 0^-} \frac{f(h + \sqrt{2}) - f(\sqrt{2})}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{2 - 2}{h} = 0$$

4.
$$f^1(0^+) = \lim_{h \rightarrow 0^+} (h \cdot \log_a a^{-1})^h \left(\frac{a^{\frac{2}{0-1-5}}}{3 + a^{1/h}} \right) = 0$$

similarly $f^1(0^-) = 0$.

$\therefore$ It is differentiable and continuous at $x = 0$

5.
$$\lim_{x \rightarrow 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + \dots + f\left(\frac{x}{k}\right) \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{f(x)}{x} + \frac{f\left(\frac{x}{2}\right)}{x} + \dots + \frac{f\left(\frac{x}{k}\right)}{x} \right)$$

$$= f^1(0) + \frac{1}{2}f^1(0) + \dots + \frac{1}{k}f^1(0)$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$$

6. (d) Continuity of $g(x)$ at $x=a$

$$\lim_{x \rightarrow a^-} g(x) = 0 \text{ and } \lim_{x \rightarrow a^+} \int_a^x f(t) dt = \int_a^a f(t) dt = 0$$

$$\text{Also, } g(a) = \int_a^a f(t) dt = 0$$

Similarly, at $x=b$ we can prove that $g(x)$ is continuous.

Thus, $g(x)$ is continuous, $\forall x \in R$.

$$\text{Now, } g'(x) = \begin{cases} 0 & , x < a \\ f(x), & a \leq x \leq b \\ 0 & , x > b \end{cases}$$

$$g'(a^-) = 0, g'(b^-) = f(b)$$

$$g'(a^+) = f(a), g'(b^+) = 0$$

$f(a)$ and $f(b)$ can never be zero as codomain of $f(x)$ is $[1, \infty)$.

Hence, $g(x)$ is non-derivable at $x=a$ and $x=b$

$$\begin{aligned} 7. \quad f(x) &= \sum_{r=1}^n \left(x^{2r} + \frac{1}{x^{2r}} + 2 \right) = \sum_{r=1}^n x^{2r} + \sum_{r=1}^n \frac{1}{x^{2r}} + 2n \\ &= (x^2 + x^4 + \dots + x^{2n}) + \left(\frac{1}{x^2} + \frac{1}{x^4} + \dots + \frac{1}{x^{2n}} \right) + 2n \\ &= \frac{x^2(1-x^{2n})}{(1-x^2)} + \frac{1}{x^2} \frac{\left(1 - \frac{1}{x^{2n}}\right)}{1 - \frac{1}{x^2}} + 2n \end{aligned}$$

$$f(x) = \frac{x^2(1-x^{2n})}{1-x^2} + \frac{(1-x^{2n})}{(1-x^2)x^2} + 2n$$

$$f(x) = \frac{(1-x^{2n})}{1-x^2} \left(x^2 + \frac{1}{x^{2n}} \right) + 2n \quad x \neq \pm 1$$

$$\therefore (f(x) - 2n)(1-x^2) = (1-x^{2n}) \left(x^2 + \frac{1}{x^{2n}} \right)$$

now consider

$$g(x) = \lim_{n \rightarrow \infty} \left((f(x) - 2n)x^{-2n-2}(1-x^2) \right) \text{ for}$$

$$x \neq \pm 1$$

$$= \lim_{n \rightarrow \infty} (1-x^{2n}) \left(x^2 + \frac{1}{x^{2n}} \right) \cdot \frac{1}{x^{2n+2}}; x \neq \pm 1$$

$$= \lim_{n \rightarrow \infty} \frac{(1-x^{2n})(x^{2n+2} + 1)}{(x^{2n})(x^{2n+2})}$$

$$= \lim_{n \rightarrow \infty} \left(-1 + \frac{1}{x^{2n}} \right) \left(1 + \frac{1}{x^{2n+2}} \right)$$

$$\text{now } g(x) = \begin{cases} -1 & \text{if } |x| > 1 \\ -\infty & \text{if } |x| < 1 \\ 0 & \text{if } x = \pm 1 \end{cases}$$

now clearly g has non removable infinite type of discontinuity at $x=1$ and -1

g is continuous at $x=2$

$$8. \quad \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 \left(a + \frac{k}{n} \right) \left(b + \frac{k}{n} \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{\left(a + \frac{k}{n} \right) \left(b + \frac{k}{n} \right)}$$

$$= \int_0^1 \frac{1}{(a+x)(b+x)} dx = \frac{1}{a-b} \int_0^1 \frac{(a+x) - (b+x)}{(a+x)(b+x)} dx$$

$$= \frac{1}{a-b} \int_0^1 \left(\frac{1}{b+x} - \frac{1}{a+x} \right) dx$$

$$= \frac{1}{a-b} \left[\ln(b+x) - \ln(a+x) \right]_0^1 = \frac{1}{a-b} \left[\ln \frac{(b+x)}{(a+x)} \right]_0^1$$

$$= \frac{1}{a-b} \ln \frac{(b+1)a}{(a+1)b}$$

if $a=b$,

$$l = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\left(a + \frac{k}{n}\right)^2} = \int_0^1 \frac{dx}{(a+x)^2} = \left[\frac{1}{a+x} \right]_0^1$$

$$= \frac{1}{a} - \frac{1}{a+1} = \frac{1}{a(a+1)}$$

9. $f(x) = \begin{cases} 5x+1, & x \leq 2 \\ \frac{x^2}{2}+1, & x > 2 \end{cases}$, so $f(x)$ is

not continuous at $x = 2$.

10. key: ABC

$$f''(x) = f'(x)$$

$$f'(x) = C_1 e^x$$

$$f(x) = C_1 e^x + C_2$$

11. (A) $f'(x) = -f'(1-x)$

$$f'(1/4) = -f'(3/4) = 0$$

$$f'(3/4) = 0$$

$$f'(1/4) = 0$$

$$\Rightarrow \exists \text{ a point } \in (1/4, 3/4) \text{ in which } f''(x) = 0$$

$$f''(1/2) = 0$$

(B) $f(x) = f(1-x)$

$$f'(x) = -f'(1-x)$$

$$f'(1/2) = -f'(1/2)$$

$$f'(1/2) = 0.$$

(C) $f(x+1/2) = f(1-x-1/2) \Rightarrow f(x+1/2) = f(1/2-x)$

i.e., $f(x+1/2)$ is even

i.e., $\int_{-1/2}^{1/2} f(x+1/2) \sin x = 0$

(D) $\int_0^{1/2} f(t) e^{\sin \pi t} dt = \int_{1/2}^1 f(1-t) e^{\sin \pi t} dt$

Put $(1-t) = u$ in RHS

$$\Rightarrow \text{R.H.S.} = - \int_{1/2}^0 f(u) e^{\sin \pi(1-u)} du = \int_0^{1/2} f(u) e^{\sin \pi u} du$$

12. $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{e^{\left(\sin \frac{\pi}{10}\right)x} f(x)}{e^{\left(\sin \frac{\pi}{10}\right)x}}$

$$= \lim_{x \rightarrow \infty} \frac{e^{\left(\sin \frac{\pi}{10}\right)x} \left(f'(x) + f(x) \sin \left(\frac{\pi}{10} \right) \right)}{e^{\left(\sin \frac{\pi}{10}\right)x} \cdot \sin \frac{\pi}{10}}$$

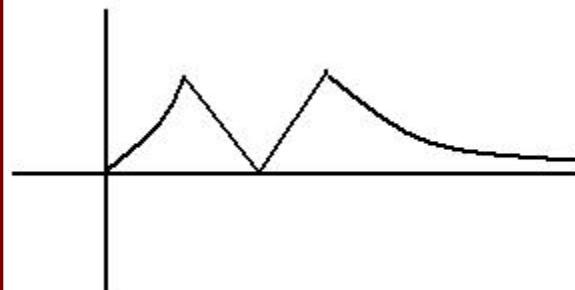
$$= \frac{\sec \left(\frac{\pi}{10} \right)}{\sin \left(\frac{\pi}{10} \right)} = \frac{4}{\sqrt{5}+1} \times \frac{4}{\sqrt{5}-1} = 4$$

13. $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{e^{\left(\tan \frac{\pi}{8}\right)\sqrt{x}} f(x)}{e^{\left(\tan \frac{\pi}{8}\right)\sqrt{x}}}$

14,15.

Setting $y = \frac{1}{x} \Rightarrow f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ as

$f(1)=2$. The graph of $h(x)$ is given below. Clearly it is not differentiable at three points.



16. (A): RHL

$$\lim_{h \rightarrow 0} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1-h^2) \right) \sin^{-1}(1-h)}{\sqrt{2}(h-h^3)}$$

$$\lim_{h \rightarrow 0} \frac{\cos^{-1}(1-h^2) \sin^{-1}(1-h)}{\sqrt{2}h(1-(h))^2}$$

$$\lim_{h \rightarrow 0} \frac{\sin^{-1} \sqrt{2h^2 - h^2} \sin^{-1}(1-h)}{\sqrt{2}h(1-(h))^2}$$

$$\lim_{h \rightarrow 0} \frac{\sin^{-1} \sqrt{2h^2 - h^2} \sin^{-1}(1-h)}{\sqrt{2}h(1-(h))^2}$$

$$\Rightarrow k = \frac{\pi}{2}$$

LHL.

$$\lim_{h \rightarrow 0} A \frac{\sin^{-1}(1-(1-h)) \cos^{-1}(1-(1-h))}{\sqrt{2(1-h)} \cdot h}$$

$$\lim_{h \rightarrow 0} A \frac{\sin^{-1} h \cos^{-1} h}{\sqrt{2(1-h)} \cdot h} = A \frac{\pi}{2\sqrt{2}}$$

$$\Rightarrow A \frac{\pi}{2\sqrt{2}} = \frac{\pi}{2} \Rightarrow A = \sqrt{2}$$

$$\therefore \sin^2 k + \cos^2 \left(\frac{A\pi}{\sqrt{2}} \right) = 1 + 1 = 2$$

$$(B): [2 + 5|n|\sin x] = 2 + [5|n|\sin x]$$

$$y = 5|n|\sin x$$

No. of points of non diff.

$$= 2(5|n| - 1) + 1$$

$$= 10|n| - 1$$

$$10|n| - 1 = 19$$

$$10|n| - 1 = 20$$

$$|n| = 2$$

$$n = \pm 2$$

(C):

$$\text{RHD: } \lim_{h \rightarrow 0} h \frac{\left(\left(\frac{3}{4} \right)^{\frac{1}{h}} - \left(\frac{3}{4} \right)^{-\frac{1}{h}} \right)}{\left(\frac{3}{4} \right)^{\frac{1}{h}} + \left(\frac{3}{4} \right)^{-\frac{1}{h}}} = -1$$

$$\text{LHD} = \lim_{h \rightarrow 0} (-h) \frac{\left(\left(\frac{3}{4} \right)^{-\frac{1}{h}} - \left(\frac{3}{4} \right)^{\frac{1}{h}} \right)}{-h} = 1$$

$$\therefore f'(0^-) - f'(0^+) = 2 = P$$

$$\text{Now } \lim_{x \rightarrow 2^-} \frac{(\exp(x+2) \ln 4)^{\frac{[x+1]}{4}} - 16}{4^x - 16} = \frac{1}{2}$$

$$17. \quad f(x) = \begin{cases} (x+2)^{x^2+\sin x}, & 1 < x < 2^{\frac{1}{3}} \\ (x+2)^{x^2+\sin x}, & 2^{\frac{1}{3}} \leq x < 3^{\frac{1}{3}} \\ (x+4)^{x^2+\sin x}, & 3^{\frac{1}{3}} \leq x < \frac{3}{2} \end{cases}$$

It is discontinuous at $x = 2^{1/3}$ and $3^{1/3}$ and hence non differentiable

$$18. \quad \text{Setting } x = y = 1 \Rightarrow f(1) = 0$$

$$f^1(x) = \lim_{h \rightarrow 0} \frac{f\left(h\left(1 + \frac{x}{h}\right)\right) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^{x\left(1+\frac{h}{x}\right)-x-1-\frac{h}{x}} \left\{ e^{1+\frac{h}{x}} f(x) + e^x \left(1 + \frac{h}{x}\right) \right\} - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x)(e^h - 1) + e^{h-1-\frac{h}{x}+x} \left\{ f\left(1 + \frac{h}{2}\right) - f(1) \right\}}{h}$$

$$= f(x) + \frac{e^{x-1} f^1(1)}{x}$$

$$\Rightarrow f^1(x) - f(x) = \frac{e^x}{x} \Rightarrow e^{-x} (f^1(x) - f(x)) = \frac{1}{x}$$

$$\Rightarrow e^{-x} f(x) = \ln|x| + c$$

$$f(1) = 0 \Rightarrow c = 0$$

$$\Rightarrow f(x) = e^x \ln|x|$$

$$f(e) = e^e$$

$$\Rightarrow \ln(f(e)) = e \Rightarrow [\ln(f(e))] = 2$$

19. We have

$$I_n = 2 \int_0^1 x \left(1 + \frac{x^2}{2} + \frac{x^4}{4} + \frac{x^6}{6} + \dots + \frac{x^{2n}}{2n} \right) dx$$

$$\left(\int_{-1}^1 (\text{odd}) dx = 0 \right)$$

$$= 2 \left[\frac{x^2}{1.2} + \frac{x^4}{2.4} + \frac{x^6}{4.6} + \dots + \frac{x^{2n+2}}{2n(2n+2)} \right]_0^1$$

$$= 2 \left[\frac{1}{1.2} + \frac{1}{2.4} + \frac{1}{4.6} + \dots + \frac{1}{2n(2n+2)} \right]$$

$$= 1 + \frac{1}{2} \left[\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \right]$$

Hence $\lim_{n \rightarrow \infty} \left[1 + \frac{1}{2} \left(1 - \frac{1}{n+1} \right) \right] = \frac{3}{2}$

$P = 3; q = 2$

SUBJECTIVE QUESTIONS

20. $f(t) = t^3 - t^2 + t + 1$

$\therefore f'(t) = 3t^2 - 2t + 1$

its disc $= (-2)^2 - 4 \cdot 3 \cdot 1 = -8 < 0$ and coefficient of $t^2 = 3 > 0$

Hence $f'(t) > 0$ for all real t . $\Rightarrow f(t)$ is always increasing

Thus $f(t)$ is maximum when t is maximum and

$t_{\max} = x$

$\therefore \max f(t) = f(x) \Rightarrow g(x) =$

$$\begin{cases} x^3 - x^2 + x + 1; 0 \leq x \leq 1 \\ 3 - x & ; \quad 1 < x \leq 2 \end{cases}$$

Now it can be easily seen that $f(x)$ is continuous in $(0, 2)$ and differentiable in $(0, 2)$ except at $x = 1$ because at $x = 1$, LHD > 0 while RHD $= -1 < 0$.

21. $\frac{dz}{dx} = \frac{1}{2} \frac{\sec^2(x/2)}{\tan(x/2)} = \frac{1}{\sin x} = \operatorname{cosec} x$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \operatorname{cosec} x \frac{dy}{dz}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dz} \left(\frac{dy}{dx} \right) \frac{dz}{dx} = \frac{d}{dz} \left(\operatorname{cosec} x \frac{dy}{dz} \right)$$

$\operatorname{cosec} x =$

$$\left(\operatorname{cosec} x \frac{d^2y}{dz^2} - \operatorname{cosec} x \cot x \frac{dx}{dz} \frac{dy}{dz} \right) \operatorname{cosec} x$$

$$= \operatorname{cosec}^2 x \frac{d^2y}{dz^2} - \operatorname{cosec} x \cot x \frac{dy}{dz}$$

Putting these values in the given differential equation, we have

$$\frac{d^2y}{dx^2} + \cot x \frac{dy}{dx} + 4y \operatorname{cosec}^2 x = 0$$

$$\Leftrightarrow \operatorname{cosec}^2 x$$

$$\frac{d^2y}{dz^2} - \operatorname{cosec} x \cot x \frac{dy}{dz} + \cot x \operatorname{cosec} x \frac{dy}{dz} + 4y \operatorname{cosec}^2 x = 0$$

$$\Leftrightarrow$$

$$\operatorname{cosec}^2 x \left(\frac{d^2y}{dz^2} + 4y \right) = 0 \Leftrightarrow \frac{d^2y}{dz^2} + 4y = 0$$

22. $g(x) = f(f(x)) = \begin{cases} 1 + f(x), & 0 \leq f(x) \leq 2 \\ 3 - f(x), & 2 < f(x) \leq 3 \end{cases}$

$$g(x) = \begin{cases} 2 + x, & 0 \leq x \leq 1 \\ 2 - x, & 1 < x \leq 2 \\ 4 - x, & 2 < x \leq 3 \end{cases}$$

Here, $g(x)$ changes the inequality sign at $x = 1$ and $x = 2$, Thus to check the continuity of $g(x)$ at $x = 1$ and $x = 2$.

At $x = 1$, L.H.L. $= \lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} (2 + x) = 3$

R. H. L. $= \lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1^+} (2 - x) = 1$

$\therefore$ L.H.L. $\neq$ R.H.L. $\Rightarrow g(x)$ is discontinuous at $x = 1$

At $x = 2$, L.H.L. $= \lim_{x \rightarrow 2^-} g(x) = \lim_{x \rightarrow 2^-} (2 - x) = 0$

R. H. L. $= \lim_{x \rightarrow 2^+} g(x) = \lim_{x \rightarrow 2^+} (4 - x) = 2$

$\therefore$ L. H. L. $\neq$ R. H. L. $\Rightarrow g(x)$ is discontinuous at $x = 2$.

23. We have $f(x) = -1, -2 \leq x \leq 0 = x - 1,$
 $0 < x \leq 2$

And $g(x) = f(|x|) + |f(x)|$

Hence $g(x)$ involves $|x|$ and $|x - 1|$ or $|-1| = 1$.

Therefore we should divide the given interval $(-2, 2)$ into the following intervals.

I_1	I_2	I_3
$[-2, 2] = [-2, 0)$	$[0, 1)$	$[1, 2]$
$x = -ve$	$+ve$	$+ve$

$$\begin{array}{lll}
 |x| = +ve & +ve & +ve \\
 f(x) = -1 & x-1 & x-1 \\
 f(|x|) = |x| - 1 & |x| - 1 = x - 1 & \\
 = -x - 1 & |x| - 1 = x - 1 & \\
 |f(x)| = |-1| = 1 & |x-1| = -(x-1) & \\
 & |x-1| = x-1 &
 \end{array}$$

$\therefore$ Using above we get $g(x) = f|x| + |f(x)|$
 $= -x - 1 + 1 = -x$ in I_1

$$= x - 1 - (x - 1) = 0$$

in $I_2, I_3, 2f'(0) = 0$

Hence $g(x)$ is defined as follows: $= 2(x - 1)$,
 $1 \leq x \leq 2$

Clearly $g'(x) = -1, 0, 2$ respectively as $g(x)$ is polynomial in x .

Also $Lg'(0) = -1$; $Rg'(0) = 0$ (not equal)

$Lg'(1) = 0$; $Rg'(1) = 2$ (not equal)

Hence $g(x)$ is not differentiable both at $x = 0$ and $x = 1$.

24. $|f(x) - f(y)| \leq |x - y|^3$ for all $x, y \in \mathbb{R} \Rightarrow$

$$\frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|^2 \text{ for all } x, y \in \mathbb{R}$$

$$\Rightarrow \lim_{y \rightarrow x} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{y \rightarrow x} |x - y|^2 \text{ for all } x$$

$$\Rightarrow \left| \lim_{y \rightarrow x} \frac{f(x) - f(y)}{x - y} \right| \leq 0 \text{ for all } x$$

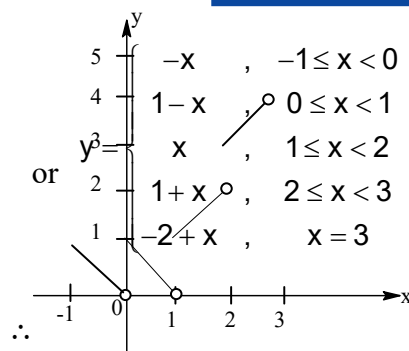
$$\Rightarrow |f'(x)| \leq 0 \text{ for all } x$$

$$\Rightarrow f'(x) = 0 \text{ for all } x$$

Hence $f(x)$ is a constant function on $\mathbb{R}$.

25.. We have, $= [x] + |1 - x|, -1 \leq x \leq 3$

$$\text{or } y = \begin{cases} -1 + 1 - x & , -1 \leq x \leq 0 \\ 0 + 1 - x & , 0 \leq x < 1 \\ 1 - 1 + x & , 1 \leq x < 2 \\ 2 - 1 + x & , 2 \leq x < 3 \\ 3 - 1 + x & , x = 3 \end{cases}$$



The graph of the given function is as shown
 From graph we can say that given function is not differentiable at $x = 0, 1, 2, 3$.

$$= \lim_{h \rightarrow 0} \frac{f(x+h) \cdot f\left(\frac{2}{x}\right) - 2f(1)}{hf\left(\frac{2}{x}\right)} = \lim_{h \rightarrow 0} \frac{2f\left(\frac{(x+h) \cdot 2}{2x}\right) - 2f(1)}{hf\left(\frac{2}{x}\right)}$$

$$= \frac{2}{f\left(\frac{2}{x}\right)} \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right) - f(1)}{\frac{h}{x} \cdot x} = \frac{2}{f\left(\frac{2}{x}\right)} \cdot \frac{f'(1)}{x}$$

$$= \frac{f(x)}{f(1)} \cdot \frac{f'(1)}{x} \Rightarrow f'(x) = \frac{f(x)}{x} \text{ \{as } f'(1) = f(1)\}$$

$$\Rightarrow xf'(x) - f(x) = 0 \Rightarrow \frac{f(x) - xf'(x)}{f^2(x)} = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{x}{f(x)} \right) = 0$$

$$\left\{ \text{as; } \frac{d}{dx} \left(\frac{x}{f(x)} \right) = \frac{f(x) \cdot 1 - x \cdot f'(x)}{f^2(x)} \right\}$$

$$\Rightarrow \frac{x}{f(x)} = \lambda \text{ (constant)} \Rightarrow \frac{f(x)}{x} = \frac{1}{\lambda} \text{ or}$$

$$f(x) = \frac{x}{\lambda} \text{ and } f(1-x) = \frac{1-x}{\lambda}$$

adding the above two we get;

$$f(x) + f(1-x) = \frac{x}{\lambda} + \frac{1-x}{\lambda} = \frac{1}{\lambda}$$

$\therefore f(x) + f(1-x) = \frac{1}{\lambda} \Rightarrow$ which is clearly constant.

(ii)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) \left[\frac{f(x+h)}{f(x)} - 1 \right]}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{f(x) \left[f\left(1 + \frac{h}{x}\right) - 1 \right]}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} f(x) \frac{\left[f\left(1 + \frac{h}{x}\right) - f(1) \right]}{x \times \frac{h}{x}}$$

$$(\because f(1) = 1)$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} \cdot f'(1) \Rightarrow f'(x) = \frac{3f(x)}{x}$$

$$\Rightarrow \log(f(x)) = 3 \log x + c$$

$$\text{at } x = 1, c = 0 \Rightarrow f(x) = x^3$$

26. $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$

$$\Rightarrow f(1) = f(1)f(1) = f^2(1)$$

$$\Rightarrow f(1) = 0 \text{ or } f(1) = 1$$

$$\Rightarrow f(1) \{f(1) - 1\} = 0$$

$$\text{If } f(1) = 0, \text{ then } f(x) = f(x \cdot 1) = f(x) \cdot f(1)$$

$$= f(x) \times 0 = 0 \text{ for all } x \in \mathbb{R}$$

Which is every where continuous and if $f(1) = 1$ It is given that f is continuous at $x = 1$, then

$$\lim_{x \rightarrow 1^+} f(x) = f(1) = \lim_{x \rightarrow 1^-} f(x)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(1+h) = f(1) = \lim_{h \rightarrow 0} f(1-h)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} f(1-h) = 1$$

Now, let a be a any nonzero real number, then

$$\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h) = \lim_{h \rightarrow 0} f\left\{a\left(1 + \frac{h}{a}\right)\right\}$$

$$= \lim_{h \rightarrow 0} f(a) f\left(1 + \frac{h}{a}\right) [\because f(xy) = f(x)f(y)]$$

$$= f(a) \lim_{h \rightarrow 0} f\left(1 + \frac{h}{a}\right) = f(a) \cdot 1 = f(a)$$

$$\text{and } \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h)$$

$$= \lim_{h \rightarrow 0} f\left\{a\left(1 - \frac{h}{a}\right)\right\}$$

$$= \lim_{h \rightarrow 0} f(a) \cdot f\left(1 - \frac{h}{a}\right) [\because f(xy) = f(x)f(y)]$$

$$= f(a) \lim_{h \rightarrow 0} f\left(1 - \frac{h}{a}\right) = f(a) \cdot 1 = f(a)$$

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = f(a) = \lim_{x \rightarrow a^-} f(x)$$

 $\Rightarrow f(x)$ is continuous at $x = a$. Since a is an arbitrary non-zero real number $\Rightarrow f(x)$ is continuous for all nonzero $x \in \mathbb{R}$.

$$27.. \text{ Given } f\left(\frac{x+y}{3}\right) = \frac{2+f(x)+f(y)}{3} \dots (1)$$

$$\therefore f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(\frac{3x+3h}{3}\right) - f\left(\frac{3x+0}{3}\right)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \left(\frac{\frac{2+f(3x)+f(3h)}{3} - \frac{2+f(3x)+f(0)}{3}}{h} \right)$$

using (1)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(3h) - f(0)}{3h} = \lim_{h \rightarrow 0} \frac{f(0+3h) - f(0)}{3h} = f'(0)$$

$$\therefore f'(x) = f'(0) = 2 \Rightarrow$$

$$\int f'(x) \cdot dx = 2 \int dx$$

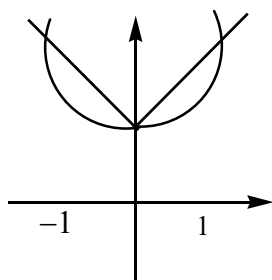
$$[\because f\left(\frac{x+y}{3}\right) = \frac{2+f(x)+f(y)}{3} \text{ putting } x=y=0]$$

$$\Rightarrow f(0) = \frac{0+2f(0)}{3} \Rightarrow f(0) = 2]$$

$$\Rightarrow f(x) = 2x + c, \text{ Putting } f(0) = c = 2$$

$$\therefore f(x) = 2x + 2$$

$$28. \quad h(x) = \begin{cases} x^2 + 1, & x \in (-\infty, -1] \\ -x + 1, & x \in [-1, 0] \\ x^2 + 1, & x \in [0, 1] \\ x + 1, & x \in [1, \infty) \end{cases}$$



$h(x)$ is not differentiable at $x = -1, 0, 1$

29. We have, $x_n^2 = a + x_{n-1}$.

It is easy to see that the variable x_n increases. Let us show that all its values remain less than some constant number. We have

$$x_{n-1}^2 - x_{n-1} - a < 0 \quad (\because x_{n-1} < x_n)$$

Hence,

$$\left(x_{n-1} - \frac{\sqrt{(4a+1)+1}}{2} \right) \left(x_{n-1} + \frac{\sqrt{(4a+1)-1}}{2} \right) < 0$$

But since the second bracketed expression exceeds by zero,

$$\text{it must be } x_{n-1} < \frac{\sqrt{4(a+1)+1}}{2}$$

$$\text{Put } \lim_{n \rightarrow \infty} x_{n-1} = \lim_{n \rightarrow \infty} x_n = \alpha$$

From the origin relation between x_n and x_{n-1} ,

$$\text{we get } \alpha^2 - \alpha - a = 0$$

$$\therefore \alpha = \frac{1 \pm \sqrt{(1+4a)}}{2}$$

$$\text{and since } \alpha \geq 0, \text{ we have } \alpha = \frac{\sqrt{(1+4a)}+1}{2}$$

30. $f(x+y) = f(x) \cdot f(y)$

$$\text{and } f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = 2$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x+0)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{f(x) \cdot f(h) - f(x) \cdot f(0)}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$\Rightarrow f'(x) = 2f(x) \Rightarrow \ln f(x) = 2x + c$$

$$\Rightarrow f(x) = e^{2x} + c \text{ at } x=0, c=0$$

$$(\because f(0)=1) \Rightarrow f(x) = e^{2x}$$

$$31. f(x) = \sqrt{|x-1|}$$

$$g(x) = \sin x$$

$$\therefore f(g(x)) = \sqrt{|\sin x - 1|}$$

$$g(f(x)) = \sin \sqrt{|x-1|} = h(x) \text{ (say)}$$

$$h'(x) = \cos \sqrt{|x-1|} \times \frac{1}{2\sqrt{|x-1|}}$$

$$\therefore h'(x) \text{ does not exist at } x=1$$

$$32. f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \text{ for all } x, y \in \mathbb{R}$$

$$f\left(\frac{x}{2}\right) = \frac{f(x)+f(0)}{2} \text{ for all } x \in \mathbb{R}$$

$$(\text{putting } y=0)$$

$$f\left(\frac{x}{2}\right) = \frac{f(x)+1}{2} \text{ for all } x \in \mathbb{R} \quad (\because f(0)=1)$$

$$\Rightarrow f(x) = 2f\left(\frac{x}{2}\right) - 1 \text{ for all } x \in \mathbb{R} \quad \dots (1)$$

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) =$$

$$\lim_{h \rightarrow 0} \frac{\left\{ 2f\left(\frac{x+h}{2}\right) - 1 \right\} - \left\{ 2f\left(\frac{x}{2}\right) - 1 \right\}}{h}$$

$$f'(x) = 2 \lim_{h \rightarrow 0} \frac{f\left(\frac{x+h}{2}\right) - f\left(\frac{x}{2}\right)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{f(x)+f(h)}{2} - f\left(\frac{x}{2}\right)}{h}$$

$$\left[\because f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x)+f(h) - 2f\left(\frac{x}{2}\right)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} \quad (\text{by (1)})$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \quad [\because f(0) = 1]$$

$$f'(x) = f'(0) \quad [\because f'(0) = -1]$$

$$\Rightarrow f'(x) = -1$$

$$\Rightarrow f(x) = -x + c \text{ for all } x \in \mathbb{R} \Rightarrow f(0) = c$$

$$\Rightarrow c = 1$$

$$\therefore f(x) = -x + 1 \Rightarrow f(2) = -2 + 1 = -1$$

33. i) $\lim_{x \rightarrow 0} \frac{2(\tan x - \sin x) - x^3}{x^5}$

$$= \lim_{x \rightarrow 0} \frac{2\left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots - x + \frac{x^3}{3} - \frac{x^5}{5}\right) - x^3}{x^5}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{2x^3}{2} + \frac{30}{120}x^5 - x^3}{x^5} = \frac{30}{120} = \frac{1}{4}$$

ii) $\lim_{x \rightarrow 1} \frac{(\ln(1+x) - \ln 2)(3 \cdot 4^{x-1} - 3x)}{[(7+x)^{1/3} - (1+3x)^{1/2}] \cdot \sin(x-1)}$

$$= \lim_{x \rightarrow 1} \frac{\ln\left(\frac{1+x}{2}\right) \cdot (4^{x-1} - x) \times ((7+x)^{1/3} + (1+3x)^{1/2})}{\sin(x-1) \times ((7+x)^{2/3} - 1 - 3x)}$$

$$= 3 \lim_{x \rightarrow 1} \frac{\ln\left(1 + \frac{x-1}{2}\right) (4^{x-1} - x) \times (2+2)}{\sin(x-1)((7+x)^{2/3} - (1+3x))}$$

$$= 12 \lim_{x \rightarrow 1} \frac{\ln\left(1 + \frac{x-1}{2}\right)}{\sin(x-1)} \times \frac{(4^{x-1} - 1 - (x-1))}{(7+x)^{2/3} - 8^{2/3} + 3 - 3x}$$

$$x - 1 = t \Rightarrow x \rightarrow 1, t \rightarrow 0$$

$$= 12 \lim_{t \rightarrow 0} \frac{\ln\left(1 + \frac{t}{2}\right)}{\sin t} \times \frac{4^t - 1 - t}{(t+8)^{2/3} - 8^{2/3} - t}$$

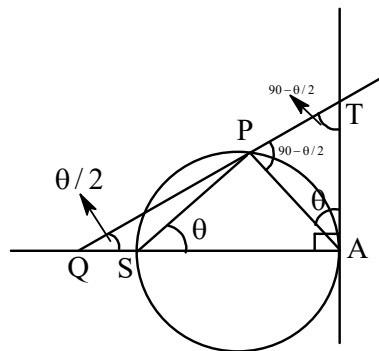
$$= 12 \lim_{t \rightarrow 0} \frac{\ln\left(1 + \frac{t}{2}\right)}{2 \times \frac{t}{2}} \times \frac{t}{\sin t} \times \frac{\frac{4^t - 1}{t} - 1}{(t+8)^{2/3} - 8^{2/3} - t}$$

$$= 12 \times \frac{1}{2} \times \frac{\ln 4 - 1}{\frac{2}{3} \cdot (8)^{\frac{2}{3}-1} - 1} = 12 \times \frac{1}{2} \times \frac{\ln(4/e)}{\frac{2}{3} \times \frac{1}{2} - 1}$$

$$= 12 \times \frac{1}{2} \times \frac{\ln 4/e \times 3}{-4/3}$$

$$= -9 \ln(4/e)$$

iii)



$$\tan\left(\frac{\theta}{2}\right) = \frac{AT}{AQ} \Rightarrow AT = AQ \tan \frac{\theta}{2} \quad \dots (1)$$

$$\sin \theta = \frac{AP}{2r}$$

$\Rightarrow AP = D \sin \theta$ (where D is the diameter of circle) ... (2)

$$\Rightarrow AQ \tan \theta/2 = 2D \sin \theta/2 \cos \theta/2$$

$$\Rightarrow AQ = 2D \cos^2 \theta/2$$

$$\lim_{\theta \rightarrow 0} AQ = 2D \lim_{\theta \rightarrow 0} \cos^2 \theta/2 \Rightarrow AQ = 2D$$

$$= 2 \times \text{Diameter of circle}$$

34. $g(x)$ is continuous at $x = \alpha$

$$\Rightarrow \lim_{x \rightarrow \alpha} g(x) = g(\alpha) \text{ and } f(x) - f(\alpha) = g(x)$$

$$(x - \alpha), x \in \mathbb{R}.$$

$$\Rightarrow \frac{f(x) - f(\alpha)}{(x - \alpha)} = g(x)$$

$$\Rightarrow \lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{(x - \alpha)} = \lim_{x \rightarrow \alpha} g(x) \Rightarrow f'(\alpha) = g(\alpha)$$

Therefore, $f(x)$ is differentiable at $x = \alpha$

Therefore, $f(x)$ is differentiable at $x = \alpha$

Conversely, f is differentiable at $x = \alpha$, then

$$\lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{(x - \alpha)} \text{ exists finitely.}$$

$$\text{Let } g(x) = \begin{cases} \frac{f(x) - f(\alpha)}{x - \alpha}, & x \neq \alpha \\ f'(\alpha), & x = \alpha \end{cases}$$

Clearly $\lim_{x \rightarrow \alpha} g(x) = f'(\alpha)$

Hence, $g(x)$ is continuous at $x = \alpha$.

Therefore $f(x)$ is differentiable at $x = \alpha$

Iff $g(x)$ is continuous at $x = \alpha$.

35. Given that $f: [-2a, 2a] \rightarrow \mathbb{R}$

f is an odd function.

$$\text{Lf' at } x = a \text{ is } 0. \Rightarrow \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{h} = 0 \quad \dots (1)$$

To find $\text{Lf' at } x = -a$ which is given by

$$\lim_{h \rightarrow 0} \frac{f(-a-h) - f(a)}{-h} = \lim_{h \rightarrow 0} \frac{-f(a+h) + f(a)}{-h}$$

$$[\because f(-x) = -f(x)]$$

$$= \lim_{h \rightarrow 0} \frac{-f(a+h) + f(a)}{-h} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Again for $x \in [a, 2a]$

$$f(x) = f(2a - x)$$

$$\therefore f(a+h) = f(2a - a - h) = f(a-h)$$

Substituting this values in last expression we get

$$\text{Lf'(-a)} = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{h} = 0$$

[Using equation (1)]

Hence $\text{Lf'(-a)} = 0$

36. To find, $\lim_{n \rightarrow \infty} \left[(n+1) \frac{2}{\pi} \cos^{-1} \left(\frac{1}{n} \right) - n \right]$

$$= \lim_{n \rightarrow \infty} n \left[\left(1 + \frac{1}{n} \right) \frac{2}{\pi} \cos^{-1} \left(\frac{1}{n} \right) - 1 \right] = \lim_{n \rightarrow \infty} n f \left(\frac{1}{n} \right)$$

$$\text{where } f(x) = \left[(1+x) \frac{2}{\pi} \cos^{-1} x - 1 \right]$$

$$f(0) = \left[(1+0) \frac{2}{\pi} \cos^{-1} 0 - 1 \right] = \frac{2}{\pi} \cdot \frac{\pi}{2} - 1 = 0$$

$$\therefore \text{Using given relation as } \lim_{n \rightarrow \infty} n f \left(\frac{1}{n} \right) = f'(0)$$

then given limit becomes

$$= f'(0) = \frac{d}{dx} \left[(1+x) \frac{2}{\pi} \cos^{-1} x - 1 \right]_{x=0}$$

$$= \frac{2}{\pi} \left[\cos^{-1} x - \frac{1+x}{\sqrt{1-x^2}} \right]_{x=0}$$

$$= \frac{2}{\pi} \left[\frac{\pi}{2} - 1 \right] = 1 - \frac{2}{\pi} = \frac{\pi-2}{\pi}$$

37. Given that,

$$f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x) \quad \dots (i)$$

$$g(x-y) = g(x) \cdot g(y) + f(x) f(y) \quad \dots (ii)$$

In equ. (i) putting $x = y$ we get

$$f(0) = f(x) g(x) - f(x) g(x) \Rightarrow f(0) = 0$$

Putting $y = 0$ in eqn. (i), we get

$$f(x) = f(x) g(0) - f(0) g(x) \Rightarrow f(x) = f(x) g(0)$$

$$[\text{using } f(0) = 0] \Rightarrow g(0) = 1$$

putting $x = y$ in eqn. (ii), we get

$$g(0) = g(x) g(x) + f(x) f(x)$$

$$\Rightarrow 1 = [g(x)]^2 + [f(x)]^2 \quad [\text{using } g(0) = 1]$$

$$\Rightarrow g(0) = 1$$

putting $x = y$ in eqn. (ii), we get $g(0) = g(x)$

$$g(x) + f(x) f(x) \Rightarrow 1 = [g(x)]^2 + [f(x)]^2$$

$$[\text{using } g(0) = 1]$$

$$\Rightarrow [g(x)]^2 = 1 - [f(x)]^2 \quad \dots (iii)$$

clearly $g(x)$ will be differentiable only if $f(x)$ is differentiable.

$\therefore$ first we will check the differentiability of $f(x)$

$$\text{Given that } \text{Rf'(0)} \text{ exists i.e., } \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

exists

$$\text{i.e., } \lim_{h \rightarrow 0} \frac{f(0)g(-h) - f(-h)g(0)}{h} \text{ exists i.e.,}$$

$$\lim_{h \rightarrow 0} \frac{-f(-h)}{h} \text{ exists}$$

(using $f(0) = 0$ and $g(0) = 1$)

which can be written as, $\Rightarrow$

$$\lim_{h \rightarrow 0} \frac{f(0) - f(-h)}{-h} = \text{Lf'(0)} \Rightarrow \text{Lf'(0)} = \text{Rf'(0)}$$

$\therefore f$ is differentiable, at $x = 0$

Differentiating equation (iii) we get

$$2g(x) \cdot g'(x) = -2f(x) \cdot f'(x)$$

for $x = 0$

$$\Rightarrow g(0) \cdot g'(0) = -f(0)f'(0) \Rightarrow g'(0) = 0$$

[using $f(0) = 0$ and $g(0) = 1$].

38. First check the differentiability:

$$\begin{aligned} Lf'(0) &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(0-h) \left\{ 1 + \frac{1}{3} \sin(\ln h^2) \right\} - 0}{(-h)} \\ &= \lim_{h \rightarrow 0} \left\{ 1 + \frac{1}{3} \sin(\ln h^2) \right\} \\ &= 1 + \frac{1}{3} \sin(\infty) = 1 + \frac{1}{3} (\text{lies between } -1 \text{ to } 1) \\ &= \text{Does not exist.} \end{aligned}$$

(Similarly $Rf'(0)$ also does not exist)

Hence $f(x)$ is not differentiable at $x = 0$.

Check the continuity at $x = 0$:

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h) \\ &= \lim_{h \rightarrow 0} f(0-h) \left\{ 1 + \frac{1}{3} \sin(\ln h^2) \right\} \\ &= 0 \left\{ 1 + \frac{1}{3} \sin \infty \right\} = 0 \end{aligned}$$

$$\begin{aligned} \text{RHL} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{x \rightarrow 0^+} f(0+h) \\ &= \lim_{h \rightarrow 0} h \left\{ 1 + \frac{1}{3} \sin(\ln h^2) \right\} \\ &= 0 \left\{ 0 + \frac{1}{3} \sin \infty \right\} = 0 \end{aligned}$$

$$\text{LHL} = \text{RHL} = 0$$

Finally $f(x)$ is continuous at $x = 0$ but not differentiable at $x = 0$.

$$\begin{aligned} 39. \quad \text{RHL} &= \lim_{x \rightarrow x^+} f(x) = \lim_{h \rightarrow 0} f(x+h) \\ &= \lim_{h \rightarrow 0} f(x) f(h) - \sqrt{6 - f(h)} \end{aligned}$$

$$\begin{aligned} &= f(x) \lim_{h \rightarrow 0} f(h) - \lim_{h \rightarrow 0} \sqrt{6 - f(h)} \\ &= f(x) \cdot 6 - 0 = 6f(x) \\ &\neq f(x) \\ &\neq \text{VF} \end{aligned}$$

This shows that if $f(x) \neq 0$, then f is

discontinuous at x . If $f(x) = 0$, then $f(x)$ is continuous at x .

$$\begin{aligned} 40. \quad \text{Given } f(x + f(y)) &= f(f(x)) + f(y) \\ \text{putting } x = y = 0 \text{ in (i), then} \\ f(0 + f(0)) &= f(f(0)) + f(0) \\ \Rightarrow f(f(0)) &= f(f(0)) + f(0) \\ \therefore f(0) &= 0 \end{aligned}$$

$$\begin{aligned} \text{Now } f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h+x) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(f(h))}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = 1 \end{aligned}$$

Integrating both sides with limits 0 to x and

$$\text{then } f(x) = x \quad \therefore f'(x) = 1$$

41. Let given limit = L , then

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left(\frac{1}{2n+1} + \frac{1}{2n+2} + \frac{1}{2n+3} + \dots + \frac{1}{4n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{2n+2} + \frac{1}{2n+4} + \frac{1}{2n+6} + \dots + \frac{1}{4n} \right) \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^{2n} \frac{n}{2n+r} - \frac{1}{n} \sum_{r=1}^n \frac{n}{2n+2r} \right] \end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^{2n} \frac{n}{2 + \frac{r}{n}} - \frac{1}{n} \sum_{r=1}^n \frac{n}{2 + 2\left(\frac{r}{n}\right)} \right] \\
&= \int_0^2 \frac{1}{2+x} dx - \int_0^1 \frac{1}{2+2x} dx \\
&= \left[\ln(2+x) \right]_0^2 - \frac{1}{2} \left[\ln(1+x) \right]_0^1 \\
&= \ln 4 - \ln 2 - \frac{1}{2} \ln 2 = \left(2 - \frac{3}{2} \right) \ln 2 \\
&= \frac{1}{2} \ln 2 = \frac{a}{b} \ln c \\
\therefore \text{Least sum } a + b + c &= 1 + 2 + 2 = 5.
\end{aligned}$$